OVERVIEW

How Transformer Explainer teaches a complex process

The page gives readers a question they can already answer in their own way: how might this sentence continue? It then lets them follow the model's answer backwards, from plausible next words to the calculations behind them. A single example ties the everyday question to the technical explanation.

One interaction shows what the design buys

I kept the opening prompt and temperature (0.8) fixed, then changed the rule for selecting the next token. The interface displayed these chances for the same five candidates.

Next token	Top-k (k = 5)	Top-p (p = 0.5)
visualize	54.10%	62.39%
create	20.65%	23.81%
see	11.96%	13.80%
make	6.19%	0%
easily	6.05%	0%

The first rule kept five candidates. The second kept enough of the highest-ranked candidates to reach a probability threshold, which in this example took three. The display redistributed the chance of selection among those three. A reader can now explain why "visualize" became more likely even though the sentence stayed the same.

Why show the alternatives?

If both runs had produced "visualize", the reader might reasonably conclude that the control did nothing. Showing the alternatives makes the effect visible even when the word drawn happens to be the same. This is the feature I would replicate first.

Scope. I inspected the desktop page, advanced the guide and tested the sampling switch on the supplied example. The model was downloading during the walkthrough, so the description of custom text generation draws on the authors' documentation. Learning outcomes below come from their published study.

Observed values from the default example. Percentages are the displayed values, rounded by the interface. Transformer Explainer · live page and illustrated article

THE LEARNING SEQUENCE

How a sentence becomes a lesson

The interface offers two routes through one diagram. A 20-card guide chooses a sequence for a newcomer. A reader with a specific question can dismiss it and inspect a component directly. The article below the diagram provides a third way to revisit the same example. The stages below group the teaching moves rather than reproduce the cards.

Reader's question	What the page does	What the reader can infer
What goes next?	The opening example is an unfinished sentence. The second card directs attention to the prompt control.	Readers can form their own expectation before they encounter unfamiliar terminology.
What travels through the model?	The five-word prompt becomes six labelled rows. The word "empowers" splits into "em" and "powers". Each piece has its own position.	A token can be part of a word. Repeating the labels later lets readers recognise the same pieces after their numerical representations change.
How does context enter?	The attention view arranges the prompt along both axes. In the expanded view, a score matrix sits beside the version with future positions masked.	Compare one row across the two views. The blocked cells show exactly which later tokens the current position cannot use.
Why so many blocks?	The overview shows one block and one attention head at a time. Stacked cards suggest other heads. Separate controls switch heads or blocks.	A head is one of several parallel views within a block. Successive blocks process the previous block's output. The two kinds of repetition have different jobs.
How does a word appear?	The output lists alternatives. Opening the probabilities view exposes the raw scores, temperature adjustment, filtering and final chances.	The model scores many possible continuations. Selecting a token is a separate step. That token can then join the prompt for the next prediction.

What the writing contributes

The prose gives each operation a job. Attention combines information across positions. The feed-forward layer, labelled MLP, then works on each position separately. That contrast explains why both appear in the same block. The web-search analogy similarly gives Query, Key and Value distinct roles. Its limit matters: attention can mix information from several positions, while an ordinary search often ends with opening one result.

EVIDENCE AND LIMITS

What the learning evidence supports

The authors randomly assigned interested, low-familiarity learners to the explainer, a blog or a video. After up to 45 minutes with their assigned resource, participants answered the same seven closed-book questions. The analysis included 90 people, 30 per group, after engagement checks excluded 38 of the original 128.

Learning resource	Average quiz accuracy
Transformer Explainer	73.3%
Selected video material	61.9%
Selected blog material	59.5%

The explainer's advantage was 11.4 percentage points over the video and 13.8 over the blog. Both comparisons reached statistical significance in the authors' planned one-sided tests ($p = 0.021$ each). Self-rated understanding differed from the blog but was similar to the video. Feeling that a resource is clear and being able to answer a question about it are separate outcomes.

The study supports the whole design for immediate conceptual understanding among motivated newcomers. It leaves open which feature caused the difference, how much people retained and how an occasional visitor would fare. Participants rated the guide particularly highly. That makes it a reasonable priority for prototyping, although a feature rating measures perceived usefulness.

Where the design asks too much of the reader

Finding a starting point. The first screen shows sampling controls, coloured paths and abbreviated labels together. The guide points to the input, but its panel also covers part of the diagram. On a smaller screen, offer the main question and result together, then open detail on request.

Reading the visual marks. Colour distinguishes query, key and value. A darker attention cell reflects a larger weight for a particular head and position. A reader could mistake it for a complete explanation of the selected word. Pair it with a local description of what the cell measures and what else contributes downstream.

Accessing the detail. Hover reveals dimensions and individual values. The loading message described a 600 MB model download and directed readers to examples. A replica should offer click and keyboard access to the same details and one complete example that works immediately. Mobile, keyboard and screen-reader usability were outside this review.

REPLICATION GUIDE

Design a first version around one comparison

Start by completing this sentence: “After using this page, a reader should be able to predict what happens to ___ when ___ changes, and explain why.” That gives the interaction a purpose and tells you what evidence to show on screen.

Worked example for an evidence atlas

Suppose a reader asks why a chart about IP activity tells a different story when it counts applicants instead of applications. Use a small, explicitly fictional dataset so the reader can check the result by hand.

Step	What to put on the screen
1 Ask	Show two regions. In North, one applicant files six applications. In South, three applicants file one each. Ask which region has more applications.
2 Predict	Show the application counts, North 6 and South 3. Before offering the toggle, ask which region would lead if each applicant counted once.
3 Change	Let the reader switch the counting unit. The same records now give North 1 and South 3. Keep the earlier totals visible alongside the new ones.
4 Explain	Let the reader expand North’s count of 1. Highlight the six records sharing its applicant ID. Add one sentence explaining how those records became one count.
5 Transfer	Add a seventh North application from the same applicant. Ask which totals will change. Success means predicting seven applications and one applicant, and explaining why.

What to build and test first

Build one complete example, one reversible control, a visible before-and-after result and one way to inspect its calculation. A labelled, precomputed example is sufficient for this first test. Add free input or animation when it answers a question the example cannot.

Give newcomers the prototype without coaching. Record whether they find the first action, predict the direction of change and explain it on a fresh example. Compare their answers with those from readers given the same content as text and a static table. If people operate the toggle but cannot explain the reversal, revise the link between records and totals before adding more controls.